

TABLE 5
ESTIMATION OF GRAIN PRODUCTION FUNCTION IN CHINA, 1952–77
Dependent Variable: ln(Grain Output)

EXPLANATORY VARIABLE	FIXED-EFFECT OLS			FIXED-EFFECT INSTRUMENTAL VARIABLES	
	(1)	(2)	(3)	(4)	(5)
ln(sown area)	.206** (.104)	.448*** (.088)	.479*** (.088)	.474*** (.092)	.442*** (.087)
ln(% acreage irrigated)		.131*** (.029)	.129*** (.027)	.127*** (.025)	.110*** (.025)
ln(% acreage sown with grain)	.019 (.059)	-.114 (.065)	-.059 (.083)	-.055 (.097)	-.091 (.092)
ln(fertilizer)	.012* (.007)	.033*** (.011)	.022** (.010)	.019* (.010)	.025** (.010)
ln(farm capital)	.245*** (.030)	.224*** (.027)	.160*** (.024)	.138*** (.030)	.158*** (.028)
ln(labor)	.578*** (.107)	.532*** (.125)	.311*** (.087)	.284*** (.088)	.343*** (.100)
ln(food availability)			.180*** (.050)	.267*** (.077)	
Tercile 1 (low)					.271*** (.108)
Tercile 2 (middle)					.246* (.137)
Tercile 3 (high)					.332** (.167)
ln(steel production)			-.093*** (.027)	-.099*** (.028)	-.090*** (.028)
ln(communal dining)		-.083 (.073)	-.045 (.031)	-.076** (.037)	-.034 (.039)
ln(production unit size)		.008* (.005)	.014** (.007)	.013** (.006)	.012** (.005)
No exit (de jure)		-.006 (.022)	-.026 (.019)	-.024 (.018)	-.038 (.023)
Good weather		-.016 (.021)	.006 (.022)	.011 (.017)	-.014 (.026)
Average weather		-.048** (.020)	-.036 (.023)	-.034* (.020)	-.044* (.025)
Bad weather		-.116*** (.027)	-.080*** (.026)	-.076*** (.023)	-.081*** (.023)
Very bad weather		-.169*** (.031)	-.161*** (.033)	-.156*** (.036)	-.158*** (.035)
Time trend		-.107*** (.042)	-.018 (.033)	-.002 (.037)	-.014 (.034)
R ²	.704	.764	.805	.800	.791
Observations	624	551	428	406	406

NOTE.—Numbers in parentheses are robust standard errors adjusted for clustering on year.

* Statistically significant at the 10 percent level.

** Statistically significant at the 5 percent level.

*** Statistically significant at the 1 percent level.